


# Engineering Electromagnetics
## 工程电磁场

Youwei Jia **嘉有为**

# Background

- 这是一门专业（微电子、通信、光电子）必修课 (Compulsory course), So you are suggested to work very hard on it.
- Total Credits: 3 (Theory: 2, Experiment: 1)
- Theory Courses: Tue 04:20-06:10 pm
- Experiment Courses: Tue (Start from 5th week), 07:00pm-8:50 pm
- Assessment of your Performance:
  Assignment: 15 % + Usual Performance 5 %
  Experiment Reports: 15 %
  Midterm Exam: 25% Final Exam: 40 %

# Background

- Part I: Vector Analysis矢量分析; Static Electric Field静电场; Steady Magnetic field稳恒磁场; Time-varying fields时变场 and Maxwell's Equations麦克斯韦方程组

  Instructor: Dr Youwei Jia

  Period: 2nd~ 15th week

Future Course: 天线与电波传播

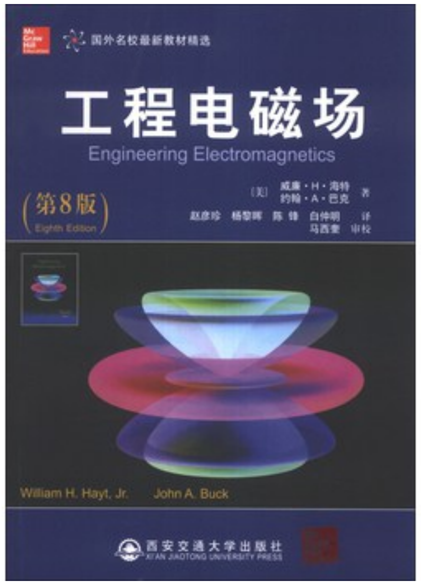
# Background

- References:

William H. Hayt and John A. Buck, 《Engineering Electromagnetics》 Mc Graw Hill, 清华大学出版社

William H. Hayt and John A. Buck, 赵彦珍等译, 《工程电磁场》 西安交通大学出版社

倪光正,《工程电磁场原理》, 高等教育出版社


# Background

- Some messages for you (Suggestions & expectations):

This course involves a lot of mathematics and physics, If you try to memorize them mechanically, it will be tough and boring. If you try to understand the natural essence 自然本质 behind the notations符号and equations公式, it will become more interesting.

My job is try my best to help you understand the most fundamental concepts概念, ideas想法, principles原理, laws规则 and theories理论. Based on that, you are expected to learn how to learn.

Keep thinking, keep practice. 勤思考， 多练习

# 1: Vector Analysis 矢量分析

- Scalar & Vector (Quantity) 标量&矢量

*Scalar* refers to a quantity whose value (值) could be represented by a real number (positive or negative).

Examples: Temperature 温度; Length 长度; Weight 重量

*A vector* quantity has both a magnitude (量、大小、模) and a direction (方向) in space (空间) .

Examples: Force 力; Speed 速度

Notations记法: Vector---**A**, $\overrightarrow{\mathbf{A}}$ ; Scalar---A, *A*

Note:
1. Always give a second thought on the terms in red.
2. Most physical quantities (scalar or vector) have their own units (单位);
3. Space has dimensions (维度，2-D, 3-D, …n-D);

# 1: Vector Analysis 矢量分析

- Field: Scalar Field & Vector Field 场、标量场、矢量场

  Region, Space 区域、空间

  Examples: a bowl of soup, energy cube, earth, universe….



When we associate some physical quantities (scalar or vector) with every point in some region/space, it involves a field (scalar or vector).

Note:

1. Mathematically, a field (scalar or vector) could be defined as some function that connects an arbitrary origin to a general point in space.

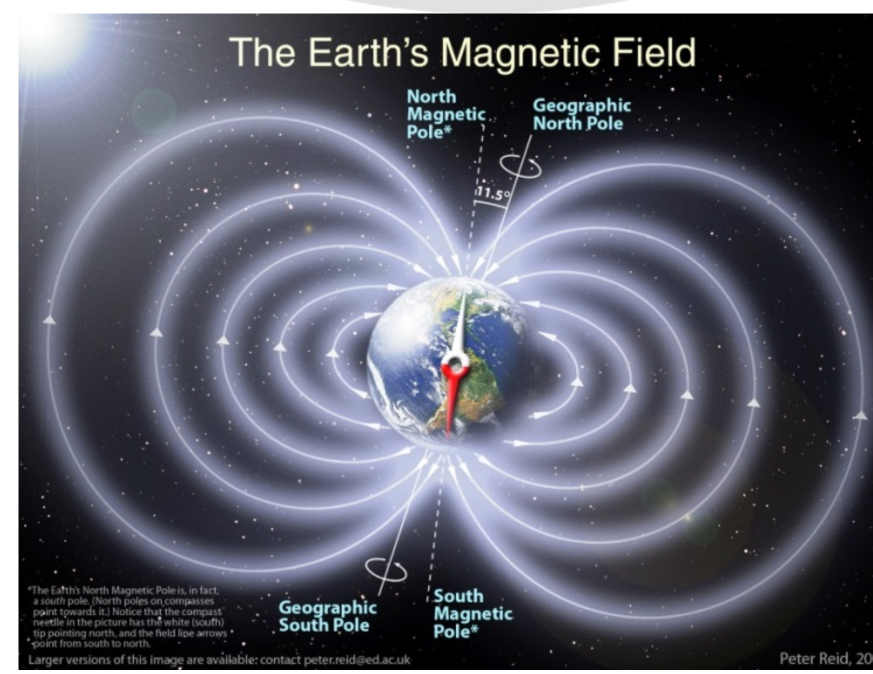
# 1: Vector Analysis 矢量分析

- Field: Scalar Field & Vector Field 场、标量场、矢量场

Examples:

- The temperature throughout the bowl of soup 温度场 (Scalar field)
  Temperature—— Scalar quantity
  The bowl of soup—— Region/Space

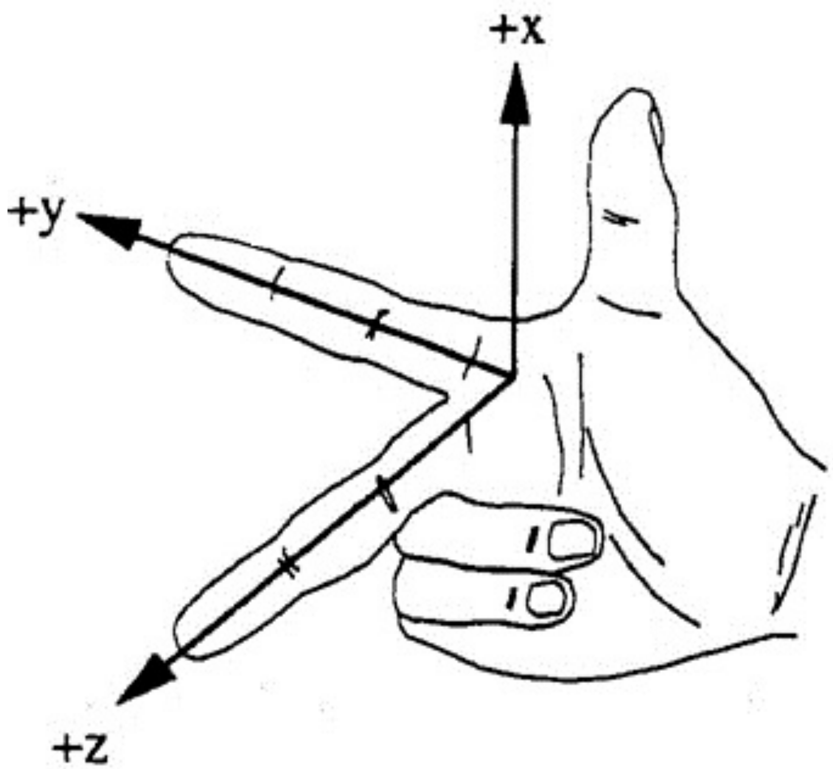
- The magnetic field of the earth 地磁场(Vector field)
  Magnetic field intensity (磁场强度)—— Vector quantity
  The earth—— Region/Space

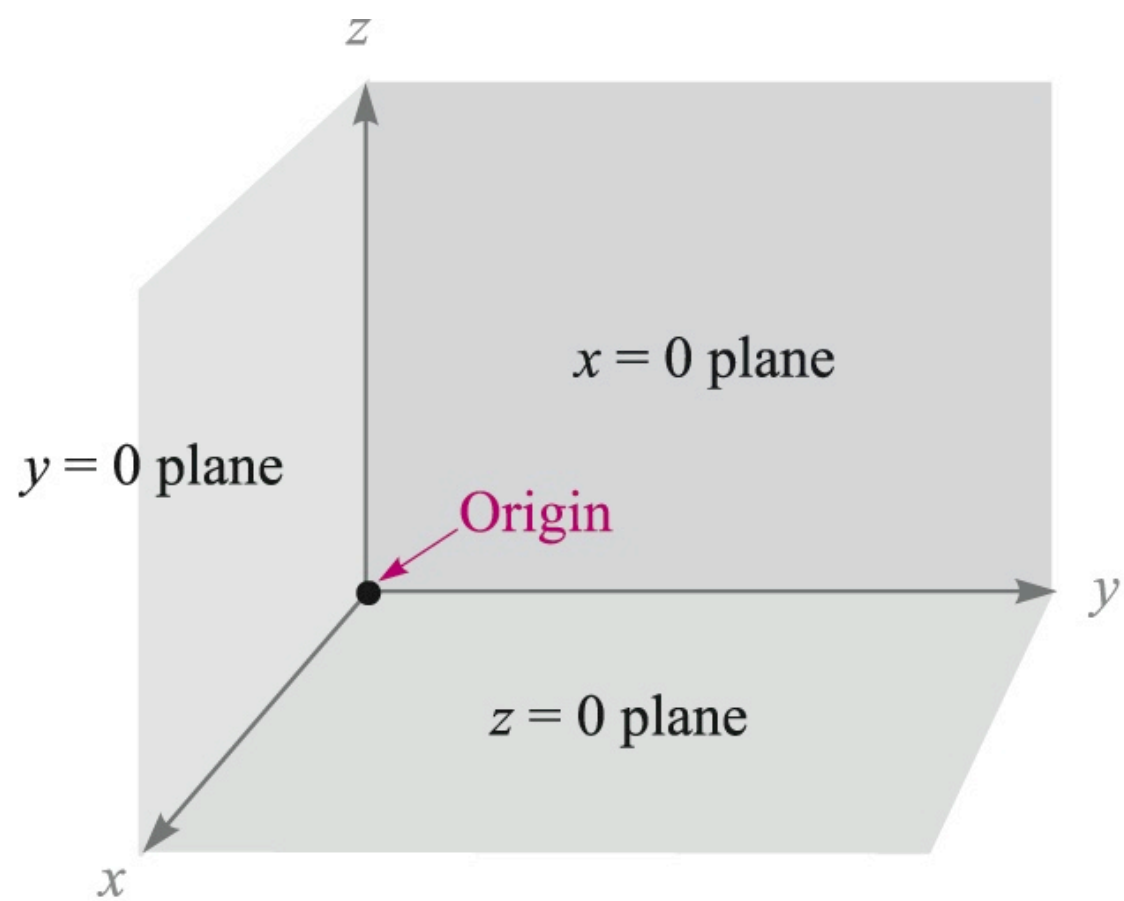


# 1: Vector Analysis 矢量分析

- The Rectangular Coordinate System 直角坐标系

——A way we define spaces, vectors



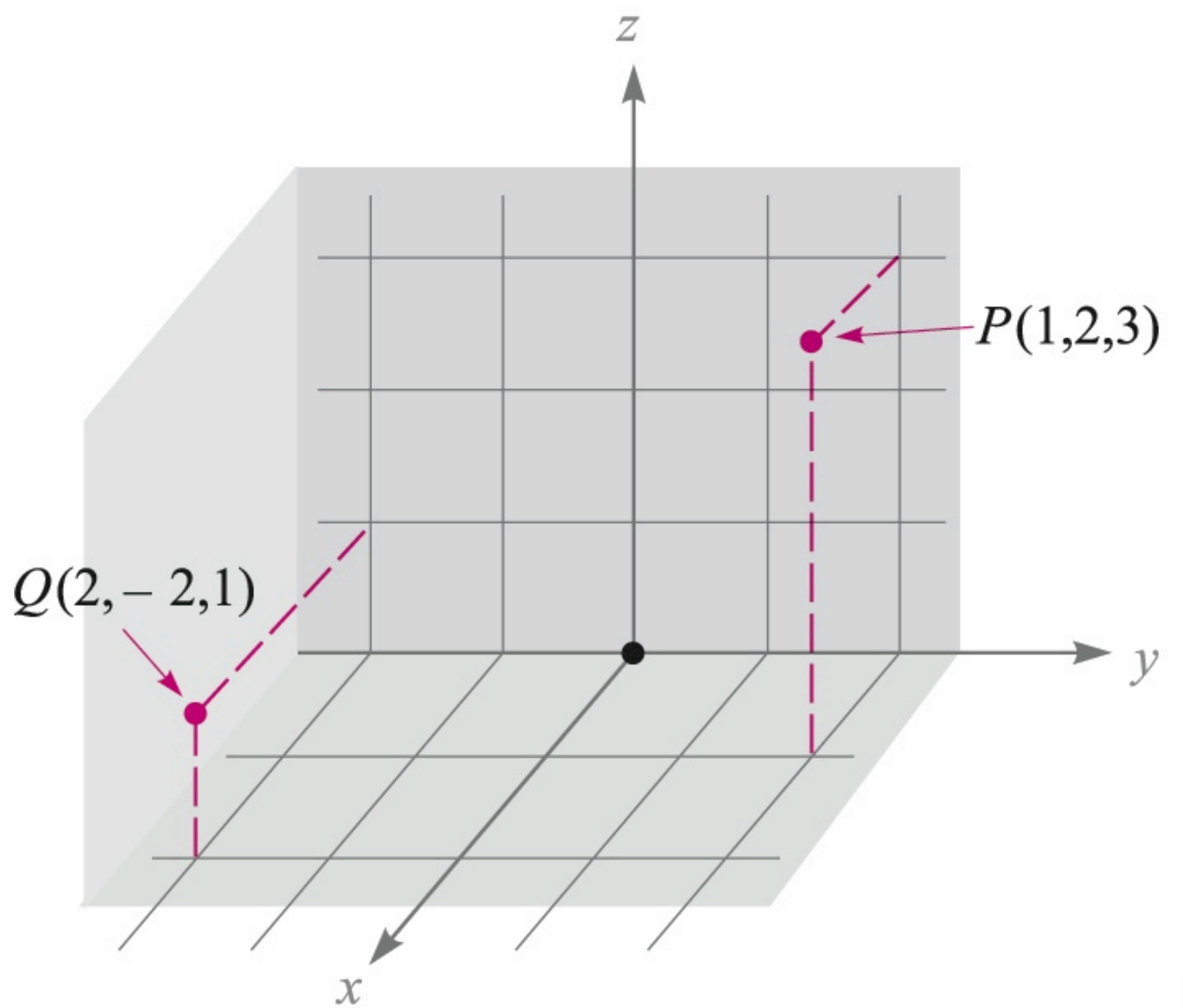

*Right-handed* coordinate system

Origin原点: Point O

Axis轴: $x$-axis, $y$-axis, $z$-axis

# 1: Vector Analysis 矢量分析

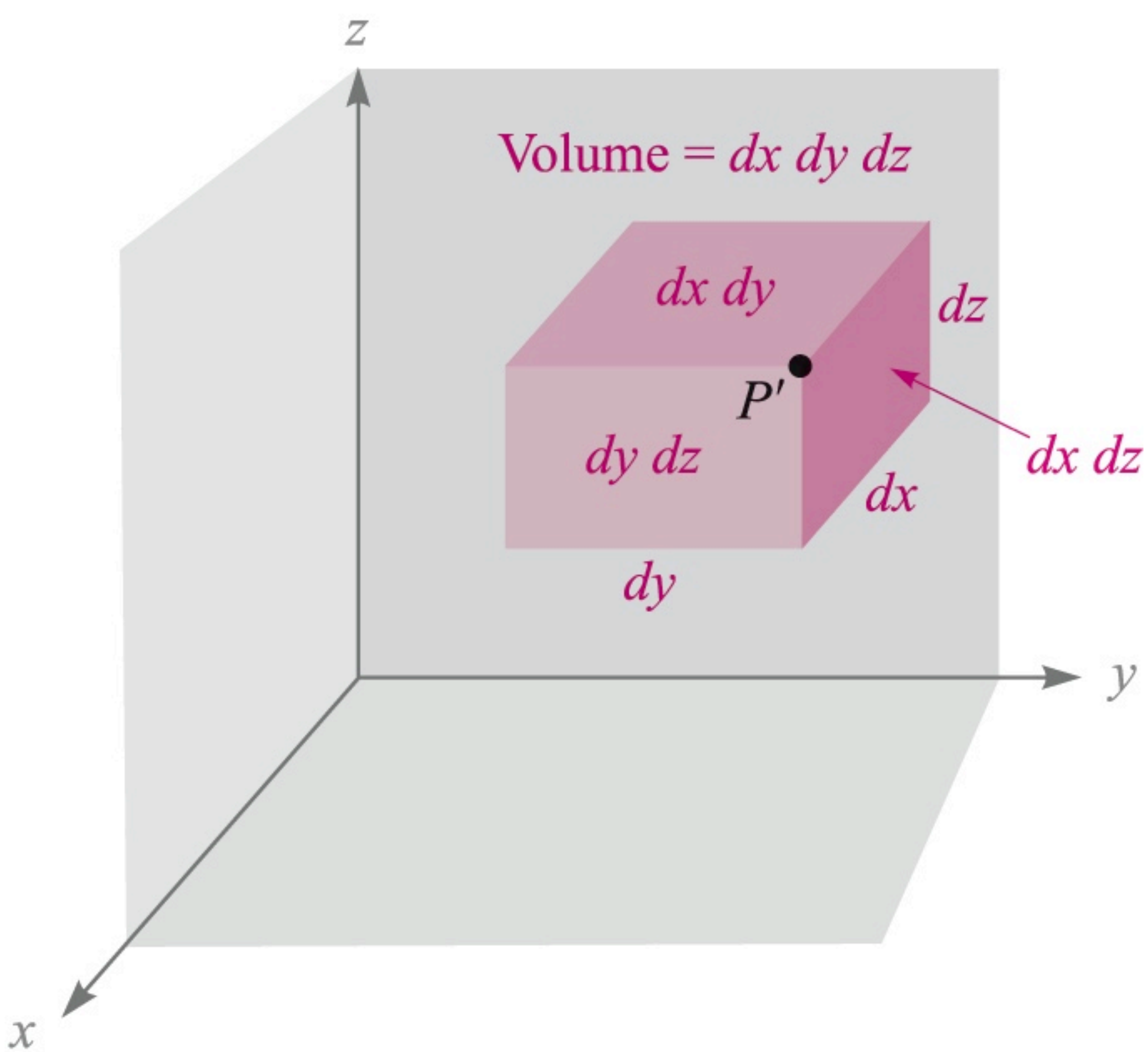
- The Rectangular Coordinate System 直角坐标系




Point locations are defined by giving their coordinates(坐标).
Examples: $P(1,2,3)$, $Q(2,-2,1)$

# 1: Vector Analysis 矢量分析

- The Rectangular Coordinate System 直角坐标系




$P(x, y, z)$

$P'(x+dx, y+dy, z+dz)$

Differential Volume Element 体微元

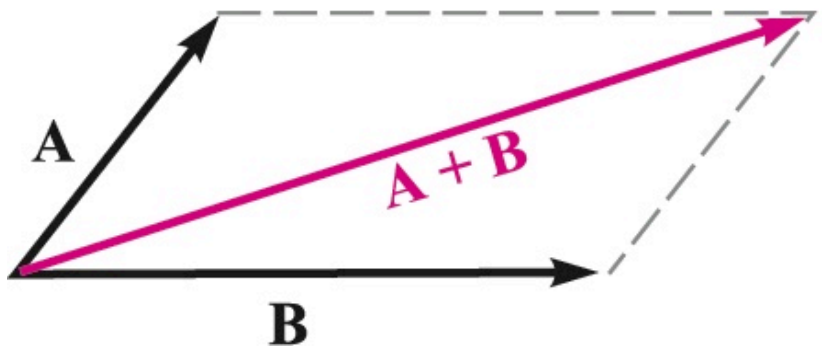
$dv=dxdydz$

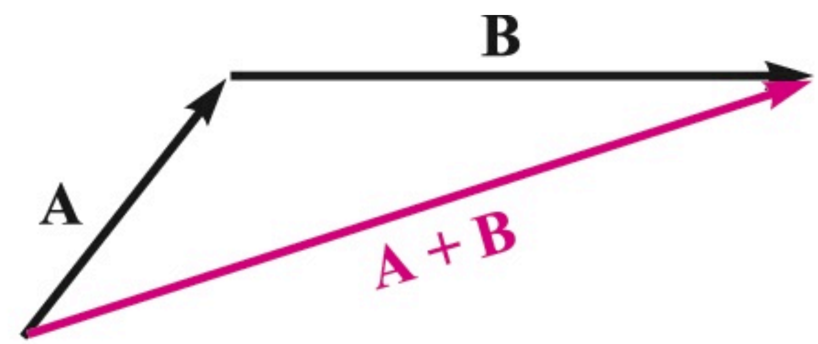
Differential Area Element 面微元

$dS=dxdy, dxdz, dydz$,

# 1: Vector Analysis 矢量分析

- Vector Addition / Subtraction 矢量加法/减法




The addition of vectors follows the parallelogram law / triangle law 平行四边形法则/三角形法则

The rule for subtraction of vectors can be easily defined from that for addition: **A-B=A+(-B)**

Associative Law 结合律:

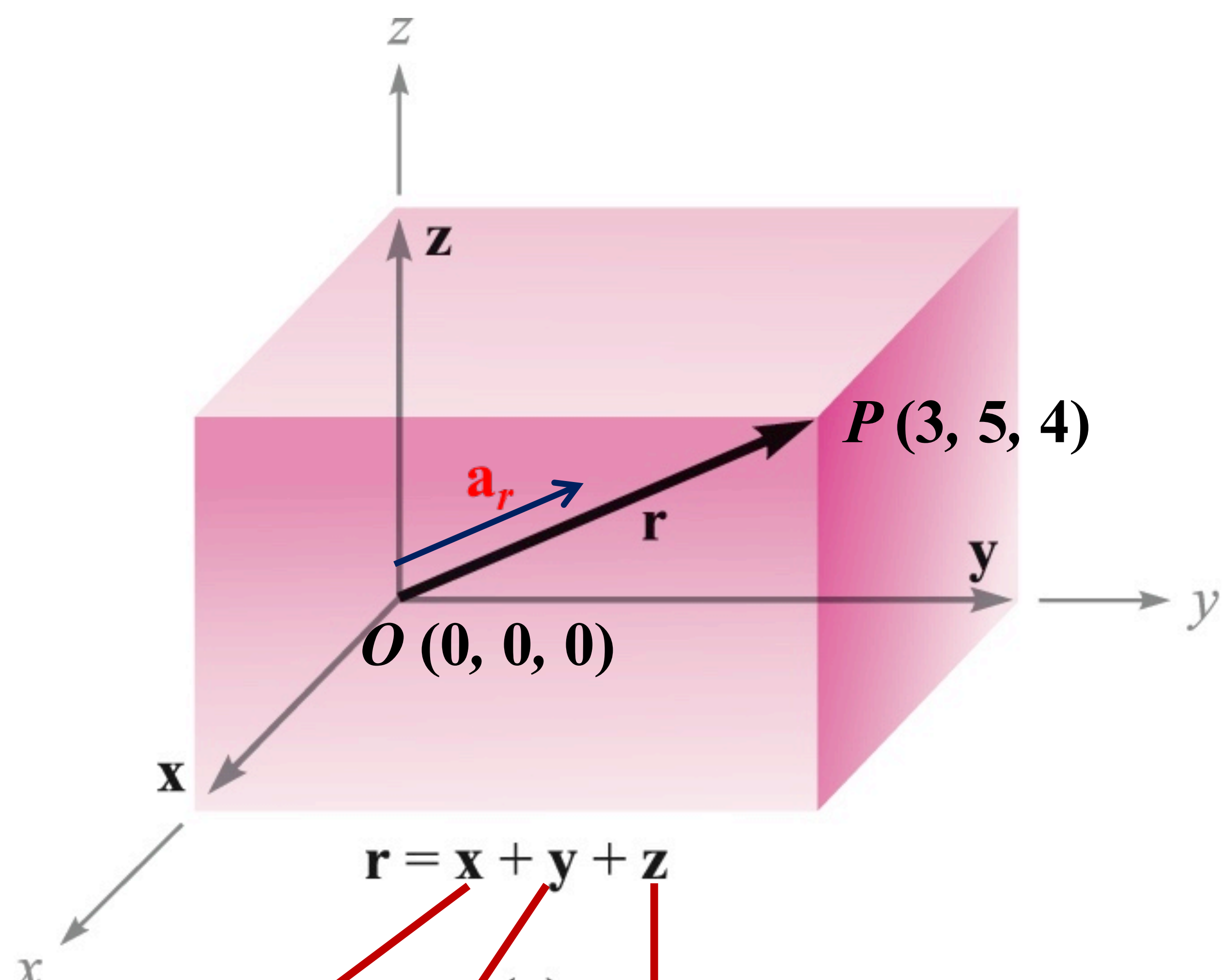
$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$$

Distributive Law分配率:

$$(r + s)(\mathbf{A} + \mathbf{B}) = r(\mathbf{A} + \mathbf{B}) + s(\mathbf{A} + \mathbf{B})$$

# 1: Vector Analysis 矢量分析

- Vector Components & Unit Vectors矢量分量&单位矢量



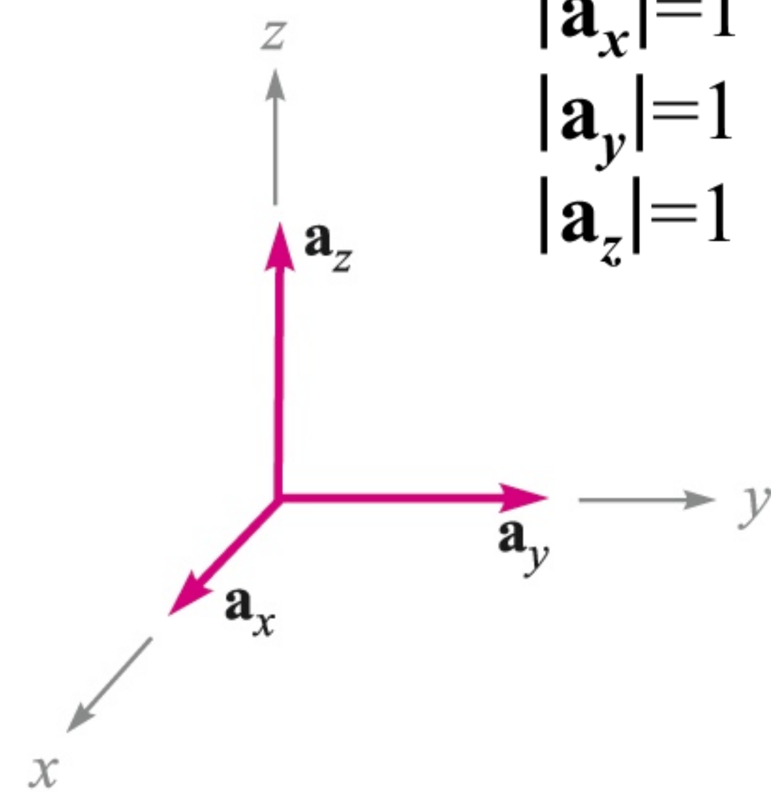

$$\mathbf{r}=3\mathbf{a}_x+5\mathbf{a}_y+4\mathbf{a}_z$$

$$|\mathbf{r}|=\sqrt{3^2+5^2+4^2}=\sqrt{50}$$

$|\mathbf{a}_x|=1$

$|\mathbf{a}_y|=1$

$|\mathbf{a}_z|=1$

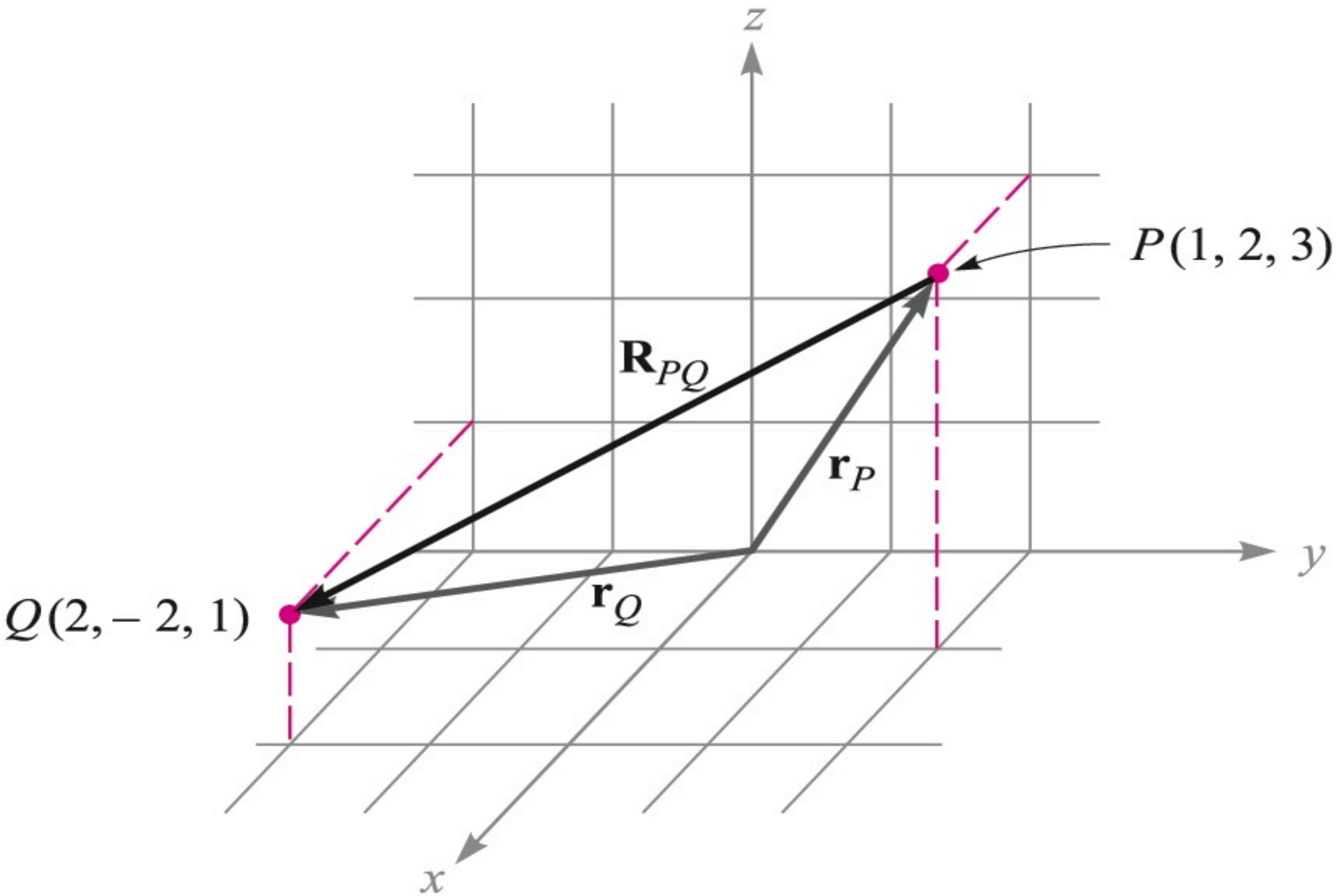



Unit Vectors

$$\mathbf{a}_r=\frac{\mathbf{r}}{|\mathbf{r}|}=\frac{3\mathbf{a}_x+5\mathbf{a}_y+4\mathbf{a}_z}{\sqrt{50}}$$

# 1: Vector Analysis 矢量分析

- Vector Components & Unit Vectors矢量分量&单位矢量




$$\mathbf{R}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P = (2-1)\mathbf{a}_x + (-2-2)\mathbf{a}_y + (1-3)\mathbf{a}_z$$
$$= \mathbf{a}_x - 4\mathbf{a}_y - 2\mathbf{a}_z$$

# 1: Vector Analysis 矢量分析

- Summary-Vector expressions in rectangular coordinates

General Vector, **B**:

$$\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

Magnitude of **B**:

$$|\mathbf{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2}$$

Unit Vector in the Direction of **B**:

$$\mathbf{a}_B = \frac{\mathbf{B}}{\sqrt{B_x^2 + B_y^2 + B_z^2}} = \frac{\mathbf{B}}{|\mathbf{B}|}$$

# 1: Vector Analysis 矢量分析

- The Dot Product of Vectors矢量点乘

Given two vectors **A** and **B**, the *dot product* is defined as the product of the magnitude of **A**, the magnitude of **B**, and the cosine of the smaller angle between them,

$$\boxed{\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}|\,|\mathbf{B}| \cos \theta_{AB}}$$

Commutative Law交换律:

$$\boxed{\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}}$$

Note: The dot product of two vectors is a scalar.

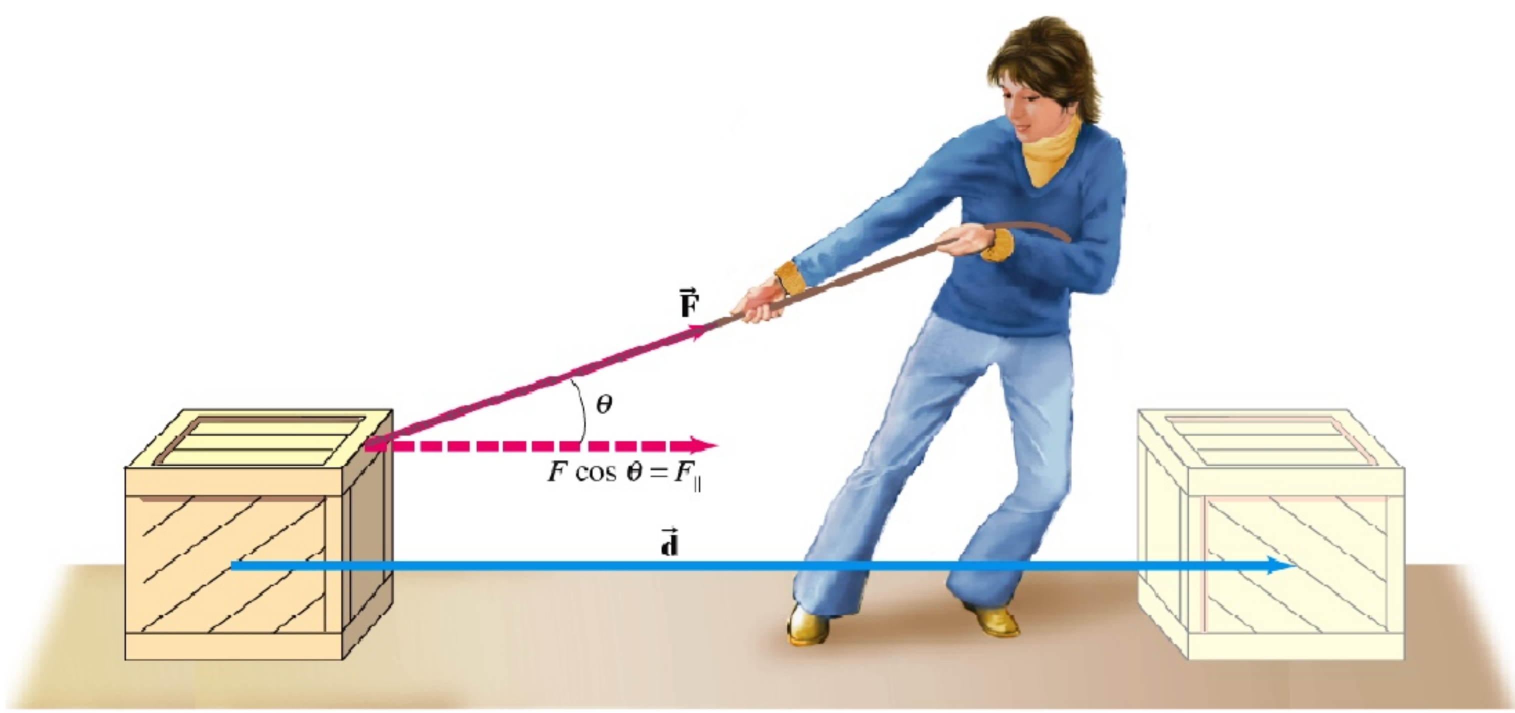
# 1: Vector Analysis 矢量分析

- The Dot Product of Vectors矢量点乘

Example: How to calculate the work (功) applied on the wood box?

$\vec{F}$: Force 力 $\vec{d}$: Displacement 位移

Work 功: $W = \vec{F} \cdot \vec{d} = |\vec{F}||\vec{d}|cos\theta$

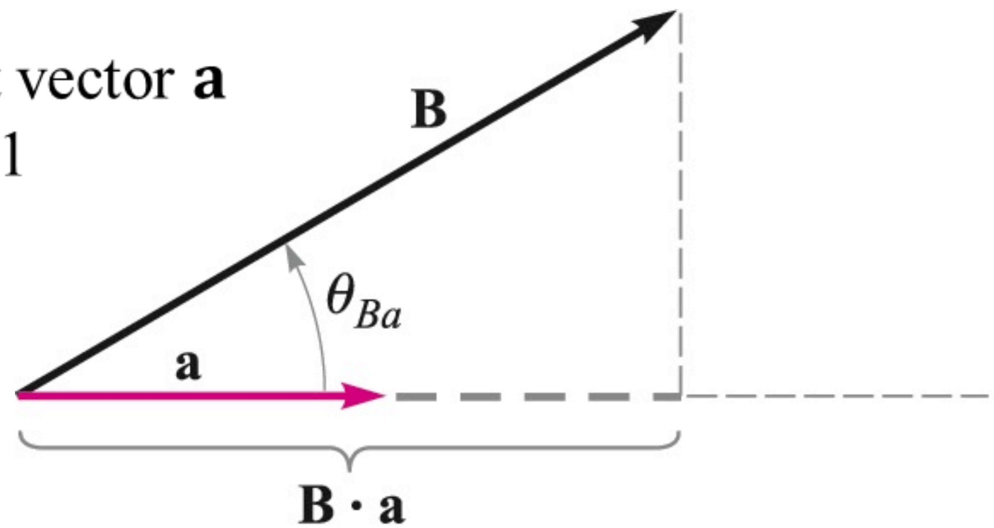


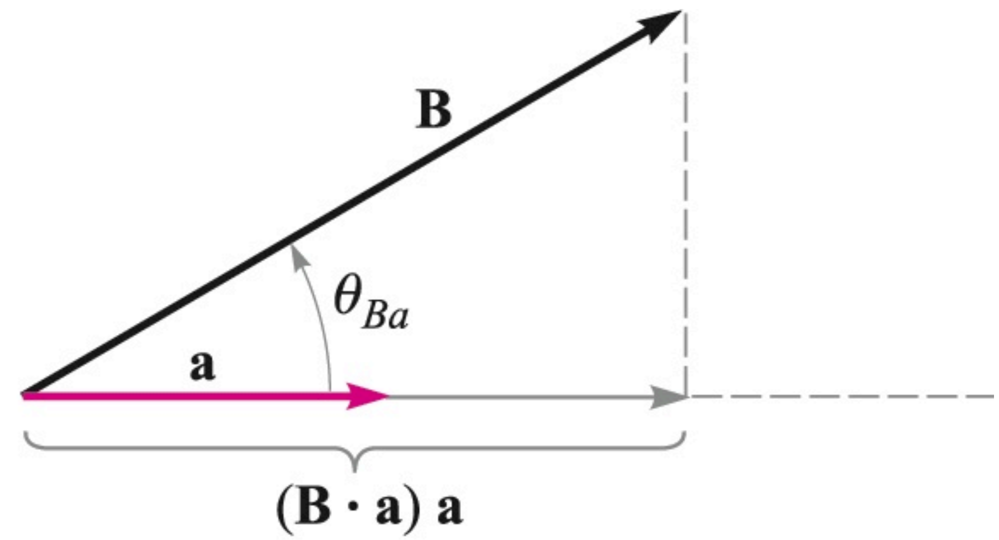
# 1: Vector Analysis 矢量分析

- The Dot Product of Vectors矢量点乘

Vector Projections (矢量投影) Using the Dot Product

Unit vector **a**
|**a**|=1



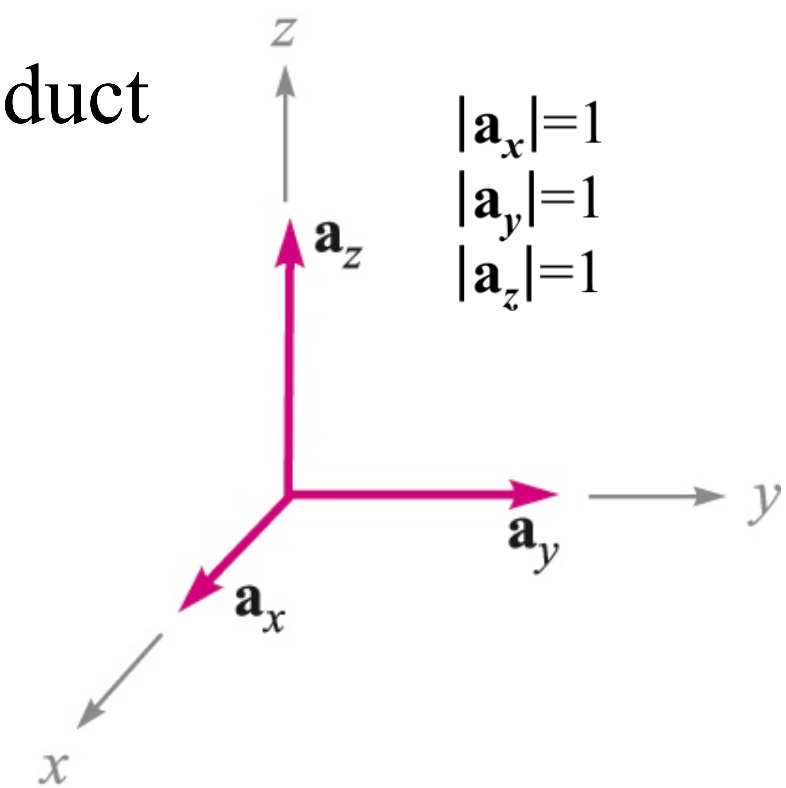

**B • a** gives the scalar component of **B** in the horizontal direction

**(B • a) a** gives the vector component of **B** in the horizontal direction

- The Dot Product of Vectors矢量点乘

Operational Use of the Dot Product

Given
$$\begin{cases} \mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z \\ \mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z \end{cases}$$




Here we use:

$$\begin{cases} \mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_z = \mathbf{a}_x \cdot \mathbf{a}_z = 0 \\ \mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1 \end{cases}$$

Finally we find

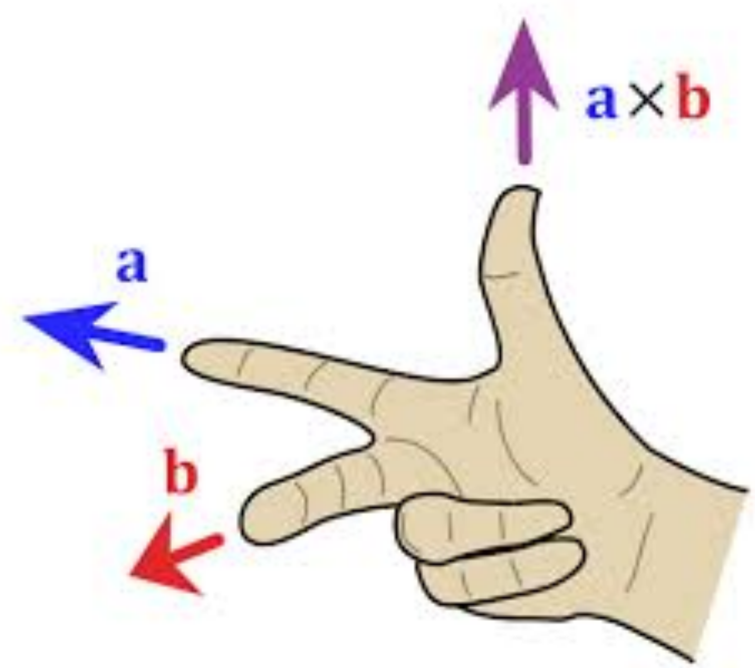
$$\boxed{\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z}$$

# 1: Vector Analysis 矢量分析

- The Cross Product of Vectors矢量叉乘

The cross product **A**×**B** is a vector; the magnitude of **A**×**B** is equal to the product of the magnitudes of **A**, **B**, and the sine of the smaller angle between **A** and **B**; the direction of **A**×**B** is perpendicular to the plane containing **A** and **B** and the right-handed law.

$$\boxed{\mathbf{A} \times \mathbf{B} = \mathbf{a}_N |\mathbf{A}|\, |\mathbf{B}| \sin\theta_{AB}}$$

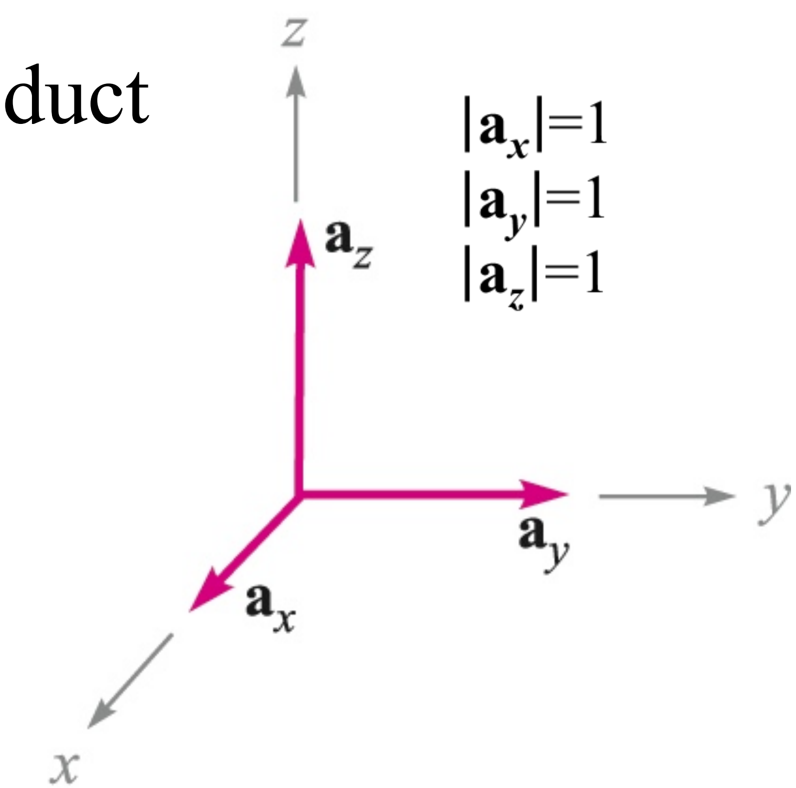


# 1: Vector Analysis 矢量分析

- The Cross Product of Vectors矢量叉乘

## Operational Use of the Cross Product

Given
$$\begin{cases} \mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z \\ \mathbf{B} = B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z \end{cases}$$



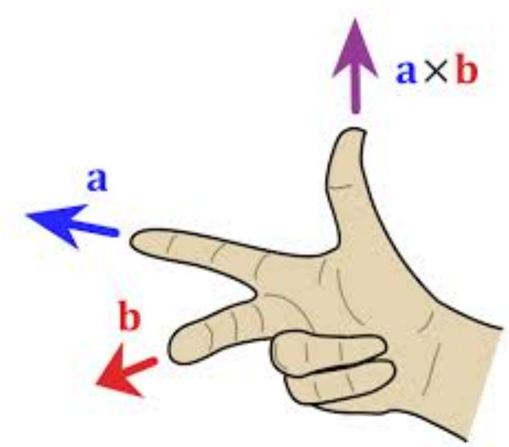

$|\mathbf{a}_x|=1$
$|\mathbf{a}_y|=1$
$|\mathbf{a}_z|=1$

$$\begin{aligned} \mathbf{A}\times\mathbf{B} = {} & A_xB_x\mathbf{a}_x\times\mathbf{a}_x + A_xB_y\mathbf{a}_x\times\mathbf{a}_y + A_xB_z\mathbf{a}_x\times\mathbf{a}_z \\ & + A_yB_x\mathbf{a}_y\times\mathbf{a}_x + A_yB_y\mathbf{a}_y\times\mathbf{a}_y + A_yB_z\mathbf{a}_y\times\mathbf{a}_z \\ & + A_zB_x\mathbf{a}_z\times\mathbf{a}_x + A_zB_y\mathbf{a}_z\times\mathbf{a}_y + A_zB_z\mathbf{a}_z\times\mathbf{a}_z \end{aligned}$$

$\mathbf{a}_x\times\mathbf{a}_x=0$ | $\mathbf{a}_x\times\mathbf{a}_y=\mathbf{a}_z$ | $\mathbf{a}_y\times\mathbf{a}_x=-\mathbf{a}_z$

$\mathbf{a}_y\times\mathbf{a}_y=0$ | $\mathbf{a}_y\times\mathbf{a}_z=\mathbf{a}_x$ | $\mathbf{a}_z\times\mathbf{a}_y=-\mathbf{a}_x$

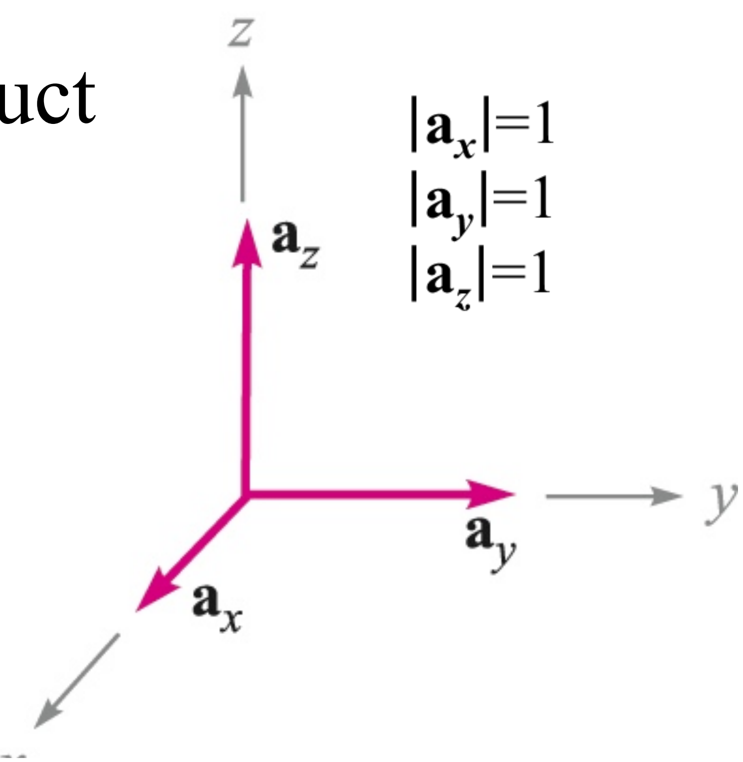
$\mathbf{a}_z\times\mathbf{a}_z=0$ | $\mathbf{a}_z\times\mathbf{a}_x=\mathbf{a}_y$ | $\mathbf{a}_x\times\mathbf{a}_z=-\mathbf{a}_y$



- The Cross Product of Vectors矢量叉乘

Operational Use of the Cross Product

Given
$$\begin{cases} \mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z \\ \mathbf{B} = B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z \end{cases}$$




Therefore:

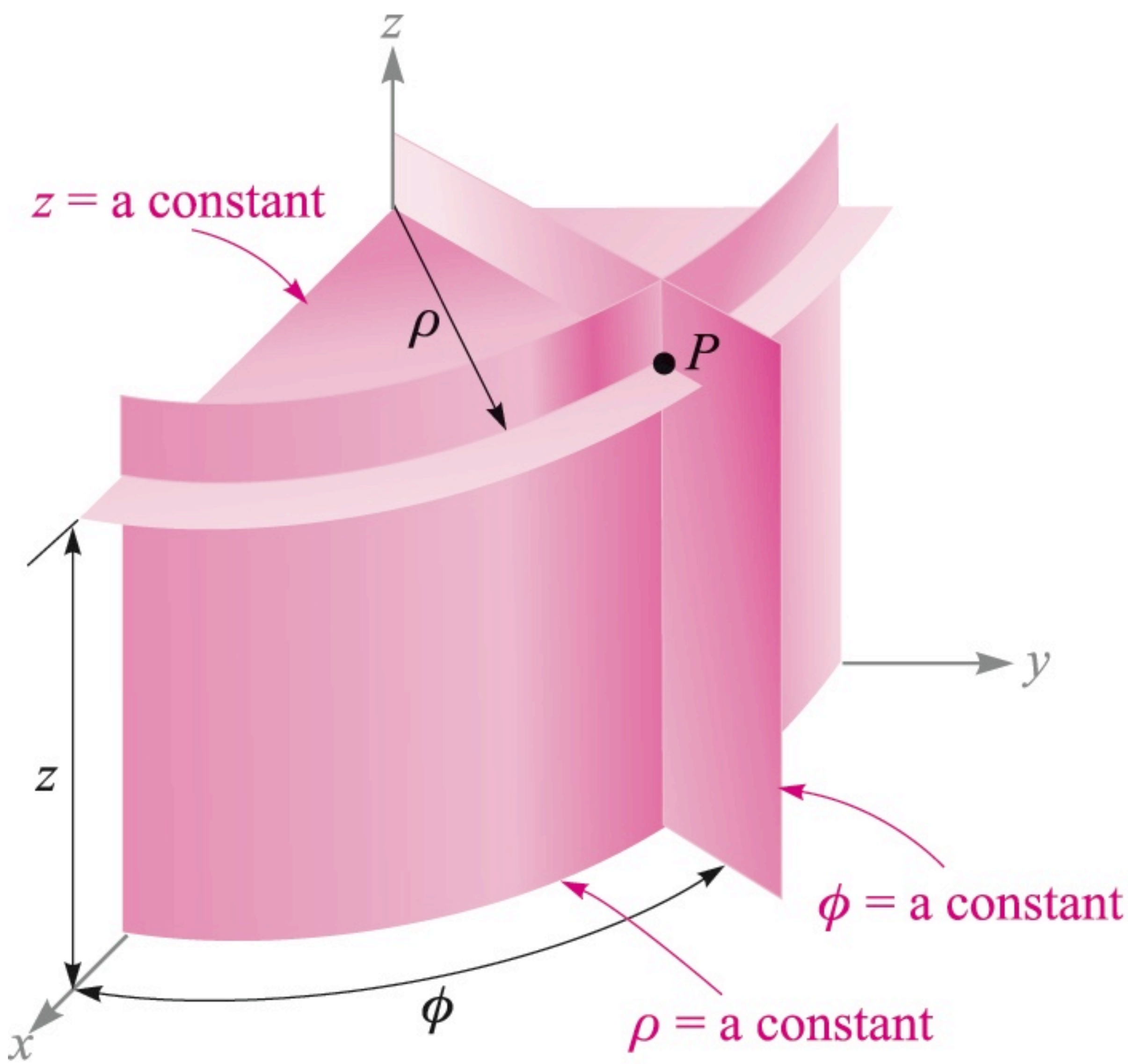
$$\mathbf{A} \times \mathbf{B} = (A_yB_z - A_zB_y)\mathbf{a}_x + (A_zB_x - A_xB_z)\mathbf{a}_y + (A_xB_y - A_yB_x)\mathbf{a}_z$$

Or…
$$\boxed{\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}}$$

A determinant in a more easily remembered form!

# 1: Vector Analysis 矢量分析

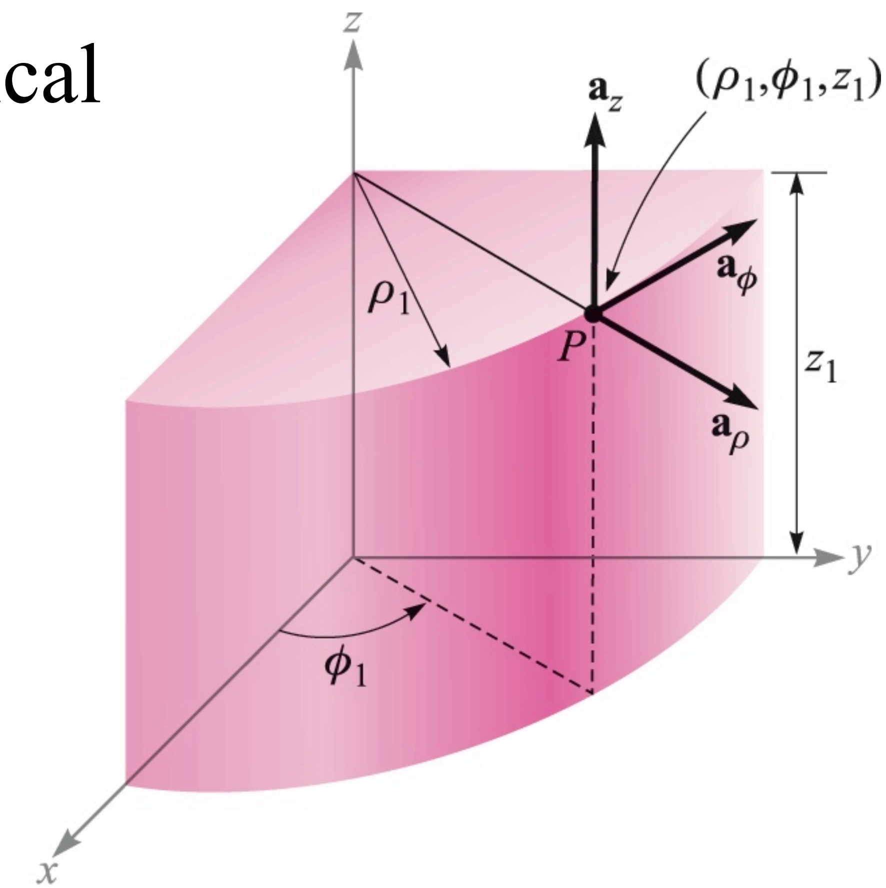
- Circular Cylindrical Coordinates圆柱坐标系




Point ***P*** has coordinates specified by ***P***($\rho,\phi,z$)

- Circular Cylindrical Coordinates圆柱坐标系

Orthogonal Unit Vectors in Cylindrical Coordinates




$$|\mathbf{a}_z| = 1$$

$$|\mathbf{a}_\rho| = 1$$

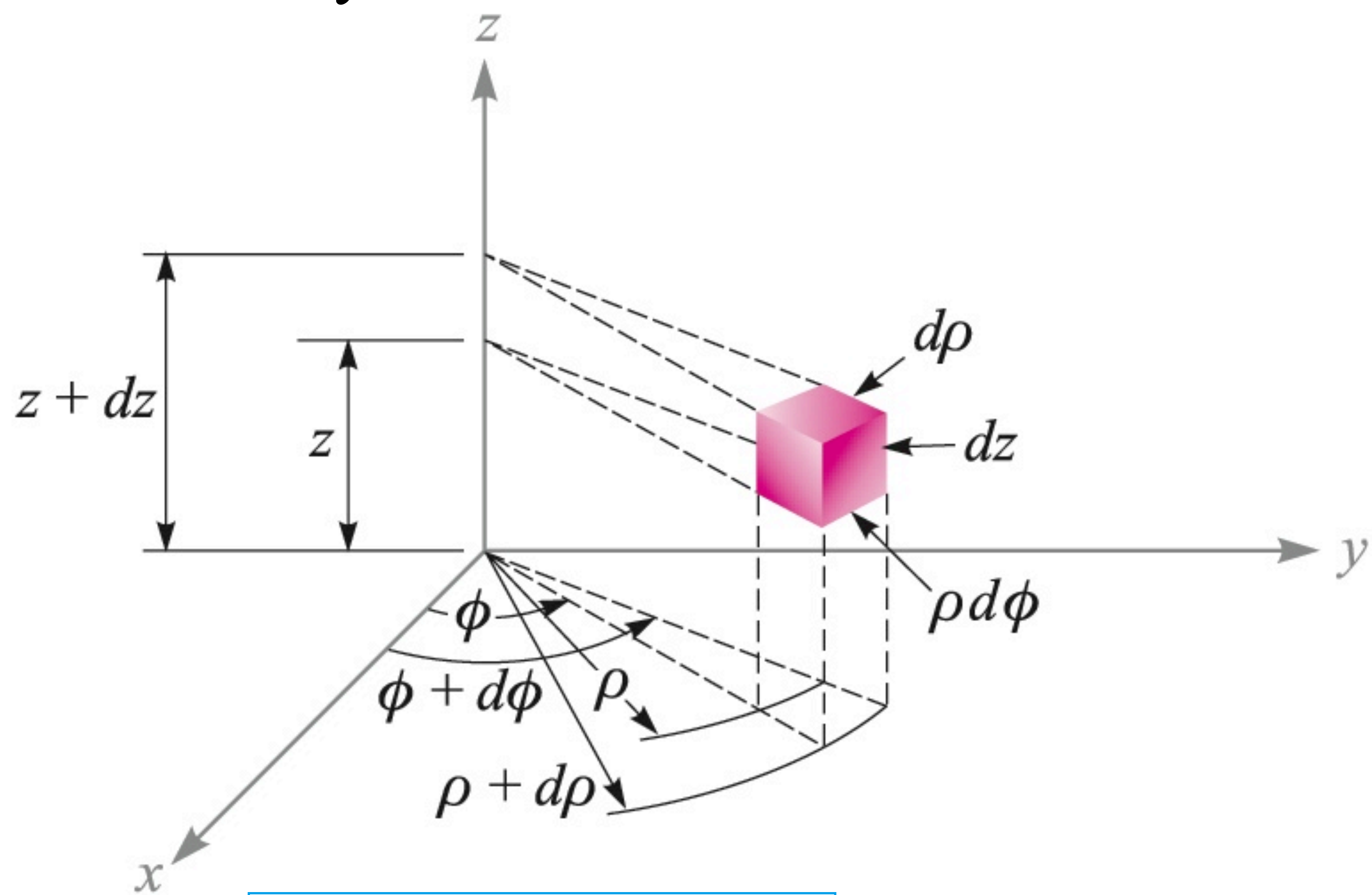
$$|\mathbf{a}_\phi| = 1$$

| | $\mathbf{a}_\rho$ | $\mathbf{a}_\phi$ | $\mathbf{a}_z$ |
|---|---|---|---|
| $\mathbf{a}_x \cdot$ | $\cos\phi$ | $-\sin\phi$ | 0 |
| $\mathbf{a}_y \cdot$ | $\sin\phi$ | $\cos\phi$ | 0 |
| $\mathbf{a}_z \cdot$ | 0 | 0 | 1 |

# 1: Vector Analysis 矢量分析

- Circular Cylindrical Coordinates圆柱坐标系

  Differential Volume in Cylindrical Coordinates




$$dV = \rho d\rho d\phi dz$$

- Circular Cylindrical Coordinates圆柱坐标系

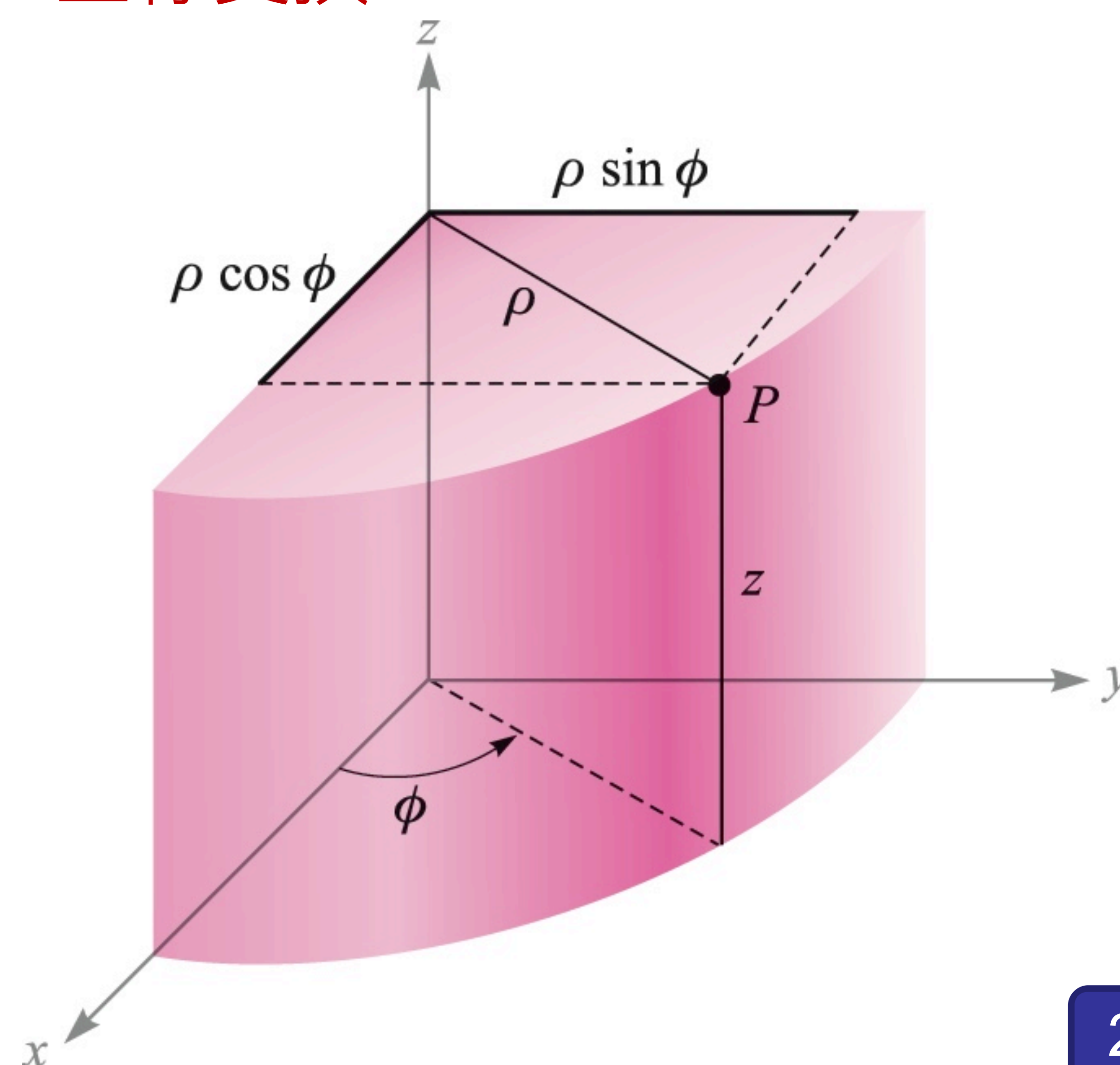
Coordinate Transformations between Cylindrical Coordinates and Rectangular Coordinates 坐标变换:

$\boldsymbol{P}(\rho,\phi,z)$

$$x = \rho\cos\phi$$
$$y = \rho\sin\phi$$
$$z = z$$

$\boldsymbol{P}(x, y, z)$

$$\rho = \sqrt{x^2 + y^2} \quad (\rho \geq 0)$$
$$\phi = \tan^{-1}\frac{y}{x}$$
$$z = z$$



- Spherical Coordinates 球坐标系

$$r = \sqrt{x^2 + y^2 + z^2} \qquad (r \geq 0)$$

$$\theta = \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}} \qquad (0^\circ \leq \theta \leq 180^\circ)$$

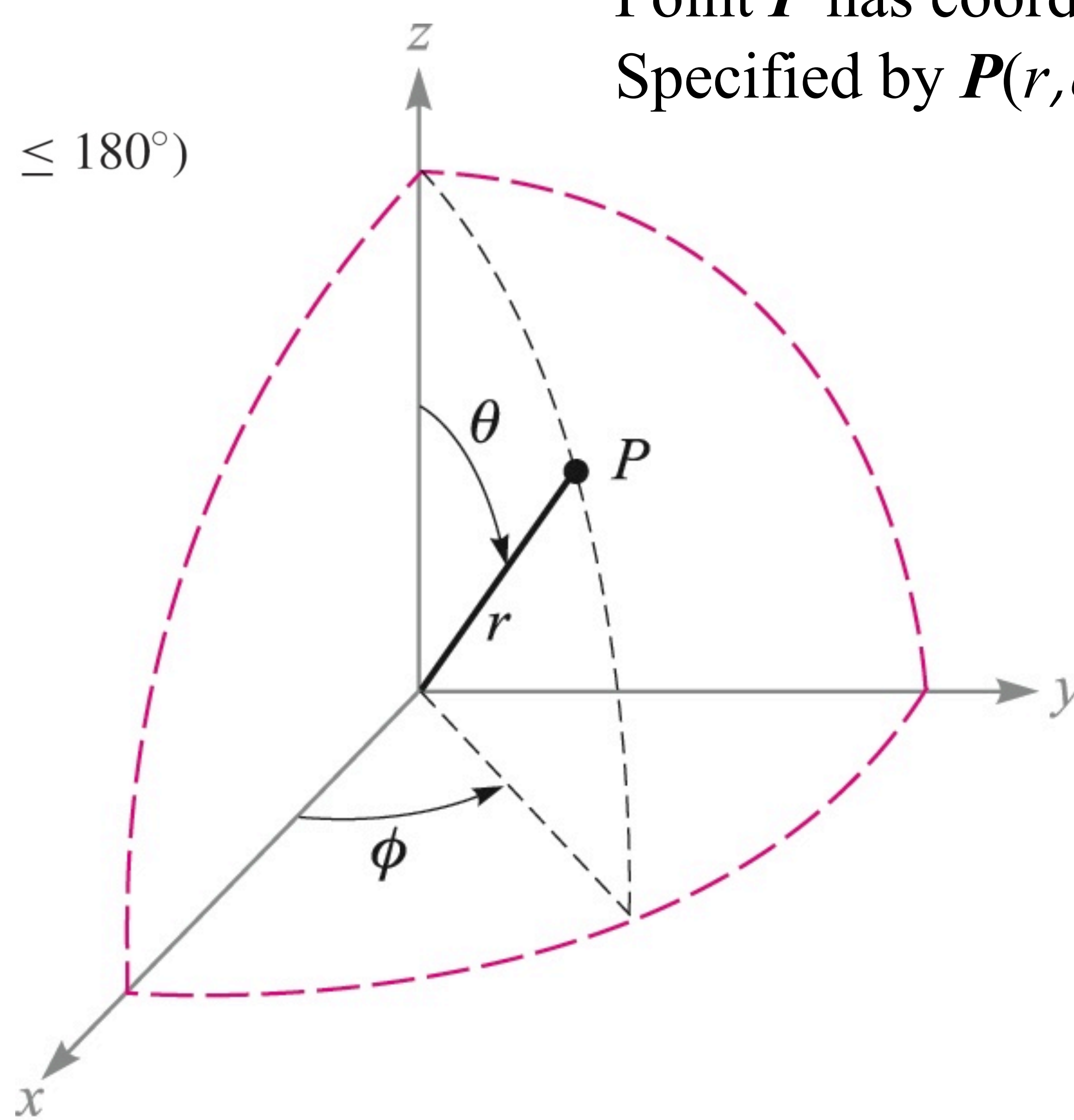
$$\phi = \tan^{-1} \frac{y}{x}$$

$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r \cos\theta$$

Point ***P*** has coordinates Specified by ***P***$(r,\theta,\phi)$



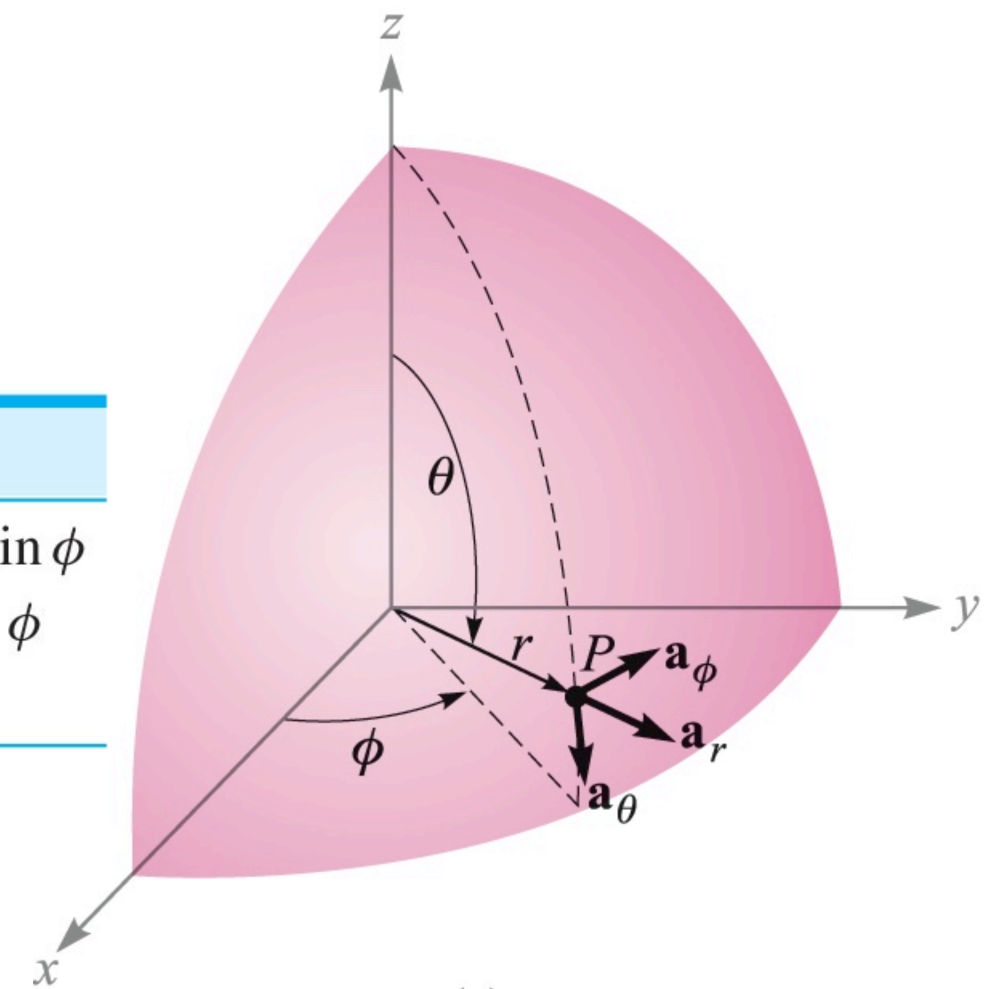
# 1: Vector Analysis 矢量分析

- Spherical Coordinates 球坐标系

Orthogonal Unit Vectors in Spherical Coordinates

$$|\mathbf{a}_r|=1 \qquad |\mathbf{a}_\theta|=1 \qquad |\mathbf{a}_\phi|=1$$

| | $\mathbf{a}_r$ | $\mathbf{a}_\theta$ | $\mathbf{a}_\phi$ |
|---|---|---|---|
| $\mathbf{a}_x\cdot$ | $\sin\theta\cos\phi$ | $\cos\theta\cos\phi$ | $-\sin\phi$ |
| $\mathbf{a}_y\cdot$ | $\sin\theta\sin\phi$ | $\cos\theta\sin\phi$ | $\cos\phi$ |
| $\mathbf{a}_z\cdot$ | $\cos\theta$ | $-\sin\theta$ | $0$ |

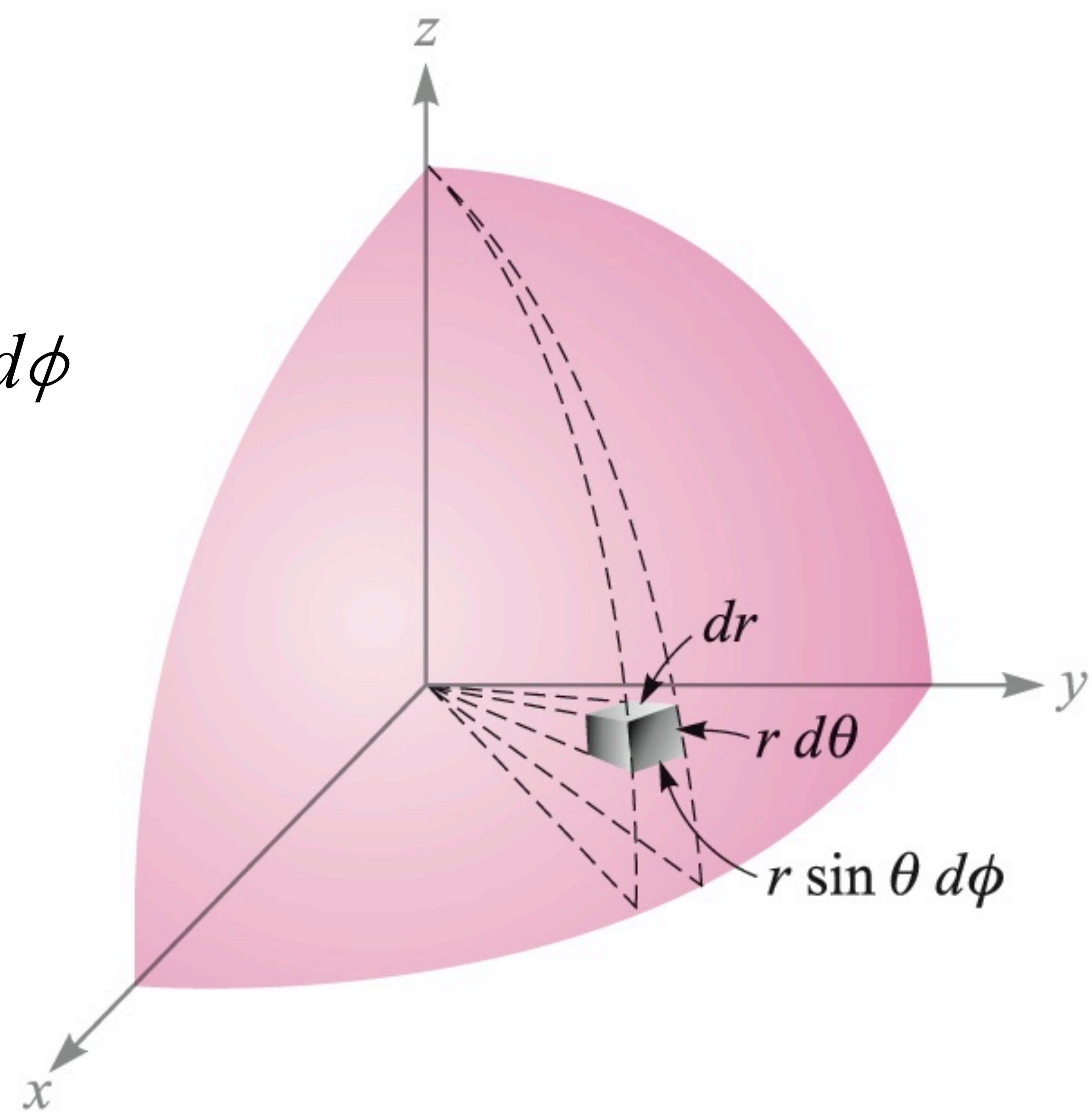


# 1: Vector Analysis 矢量分析

- Spherical Coordinates 球坐标系

Differential Volume in Spherical Coordinates

$$dV = r^2 \sin\theta dr d\theta d\phi$$



# 1: Vector Analysis 矢量分析

We illustrate this procedure by transforming the vector field $\mathbf{G} = (xz/y)\mathbf{a}_x$ into spherical components and variables.

***Solution.*** We find the three spherical components by dotting $\mathbf{G}$ with the appropriate unit vectors, and we change variables during the procedure:

$$G_r = \mathbf{G} \cdot \mathbf{a}_r = \frac{xz}{y}\mathbf{a}_x \cdot \mathbf{a}_r = \frac{xz}{y} \sin\theta \cos\phi$$

$$= r \sin\theta \cos\theta \frac{\cos^2\phi}{\sin\phi}$$

$$G_\theta = \mathbf{G} \cdot \mathbf{a}_\theta = \frac{xz}{y}\mathbf{a}_x \cdot \mathbf{a}_\theta = \frac{xz}{y} \cos\theta \cos\phi$$

$$= r \cos^2\theta \frac{\cos^2\phi}{\sin\phi}$$

$$G\phi = \mathbf{G} \cdot \mathbf{a}_\phi = \frac{xz}{y}\mathbf{a}_x \cdot \mathbf{a}_\phi = \frac{xz}{y}(-\sin\phi)$$

$$= -r \cos\theta \cos\phi$$

Collecting these results, we have

$$\mathbf{G} = r \cos\theta \cos\phi \, (\sin\theta \cot\phi \, \mathbf{a}_r + \cos\theta \cot\phi \, \mathbf{a}_\theta - \mathbf{a}_\phi)$$

# 1: Vector Analysis 矢量分析

- Coordinate Transformation

**D1.6.** Transform to cylindrical coordinates: (*a*) $\mathbf{F} = 10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z$ at point $P(10, -8, 6)$; (*b*) $\mathbf{G} = (2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y$ at point $Q(\rho, \phi, z)$. (*c*) Give the rectangular components of the vector $\mathbf{H} = 20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z$ at $P(x = 5, y = 2, z = -1)$.

# HW1

1. Review all the lecture notes, especially, Page 6, 7, 12, 14, 18, 20, 21, 23, 25, 26, 27. Try to understand the meaning of the notations and equations.

2. Vector **A** extends from the origin to (1,2,3) and vector **B** from the origin to (2,3,-2). 1) Find the unit vector in the direction of (**A** − **B**); 2) find the unit vector in the direction of the line extending from the origin to the **midpoint** of the line joining the ends of **A** and **B**.

3. The vector from the origin to the point *A* is given as (6,−2,−4), and the unit vector directed from the origin toward point *B* is (2,−2, 1)/3. If points *A* and *B* are ten units apart, find the coordinates of point *B*.

4. A vector field is specified as $\mathbf{G} = 24xy\mathbf{a}_x + 12(x^2 + 2)\mathbf{a}_y + 18z^2\mathbf{a}_z$. Given two points, *P*(1, 2,−1) and *Q*(−2, 1, 3), find: 1) **G** at *P*; 2) a unit vector in the direction of **G** at *Q*; 3) a unit vector directed from *Q* toward *P*; 4) the equation of the surface on which $|\mathbf{G}| = 60$.

5. Three vectors extending from the origin are given as $\mathbf{r}_1 = (7, 3, -2)$, $\mathbf{r}_2 = (-2, 7, -3)$, and $\mathbf{r}_3 = (0, 2, 3)$. Find: 1) a unit vector perpendicular to both $\mathbf{r}_1$ and $\mathbf{r}_2$; 2) a unit vector perpendicular to the vectors $\mathbf{r}_1 - \mathbf{r}_2$ and $\mathbf{r}_2 - \mathbf{r}_3$; 3) the area of the triangle defined by $\mathbf{r}_1$ and $\mathbf{r}_2$; 4) the area of the triangle defined by the heads of $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$. 提示：两个矢量叉乘的大小与这两个矢量张成的平行四边形的面积相等。